

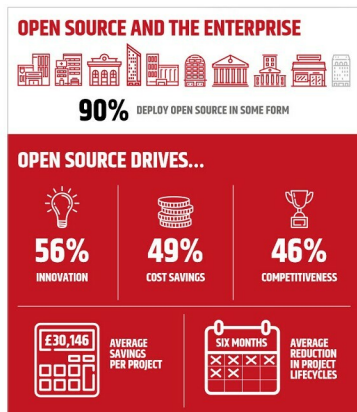


Figure 1: 2016-2017 trends in open source adoption (Image courtesy: Rackspace)

in no manner representative of relative importance.

1. Hyper: A terminal built using HTML, CSS and JS

The creators of Hyper had one basic goal in mind—to create a simple, efficient and, most importantly, hackable command line interface for developers. Intuitive shortcuts can be added in Hyper based on developer preferences, and there is the option of building your own plugins to provide additional functionality to this tiny but powerful application.

2. Parse Server: The open source version of the Parse back-end

This software can be deployed in any infrastructure that is capable of running Node.js, and works well with the Express Web Application Framework. It can be deployed independently, or added to existing Web applications. Supported by an array of tutorials and extensive documentation, this has proved to be a popular, community-backed project.

3. TensorFlow: Google's open source machine learning framework

Originally released in 2015, the TensorFlow project saw major updates in 2017. These included support for Python generators, additions to various APIs for performance improvements and, most importantly, the addition of Keras to the TensorFlow core package. In addition, Tensorboard was released to help improve visualisation via plotting of quantitative metrics on a graph. TensorFlow is currently the

most popular project on GitHub with over 77,000 stars. In comparison, the Linux project has just about 53,000 stars!

4. Caffe2 and PyTorch: Updates to both open source machine learning frameworks

Facebook-backed PyTorch was developed in response to Google's release of TensorFlow. Facebook also uses Caffe for training deep learning models in production. An extremely interesting development in the recent releases of both these frameworks is due to Facebook and Microsoft's partnership on releasing updates to these frameworks based on a new protocol defined as Open Neural Network Exchange (ONNX). ONNX is a standard for the representation of deep learning models that allows models to be transferred between frameworks. It is going to be interesting to monitor progress on this front.

5. Bulma: A FOSS CSS framework based on Flexbox

Available as a more intuitive alternative to popularly used Bootstrap, Bulma is essentially a single file—*bulma.css*; it does not include any JavaScript since people generally want to use their own JavaScript implementation. It can be considered 'environment agnostic' since it's just the style layer on top of the logic.

6. Anime: A lightweight, open source JS library tailored for animation

This is a lightweight JavaScript animation library. Anime works with any of CSS' properties, individual CSS transforms, SVG or any DOM attributes, as well as with JavaScript Objects.

7. Yarn: A fast, reliable and secure dependency management software

Introducing a number of unconventional and innovative features, Yarn presents a convenient alternative to the software in use. What it offers ranges from offline installs on account of caching, to increased reliability and security, arising from paradigms within the installation that offer network resilience and concurrency for performance. Yarn goes the distance for developers looking for more efficiency and less failures across systems resulting from mismanagement of dependencies.

8. Apache Software: Multiple projects by the Apache Software Foundation

From releases of Hadoop to Spark, the Apache Software Foundation has had a number of important releases for projects in the open source domain this year.

CarbonData is a novel Big Data file format for interactive querying using advanced columnar storage. TinkerPop is another project that works behind the scenes in powering graph-script modelling, formulation and analysis for a number